

Basic Possibilistic Logic: Examples and Visualization

Seminar Paper

submitted by

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Abstract

Possibilistic logic is a non-classical logic framework designed to handle uncertainty by incorporating degrees of possibility. This paper provides a general introduction to possibilistic logic by using strong examples of the application.

1 Introduction

Reasoning under uncertainty is a significant problem in artificial intelligence. Several approaches have been proposed to address this issue, including possibility theory [8], fuzzy logic [13], and possibilistic logic [1]. The foundation of possibility theory was laid by Zadeh in 1978 [14] as an extension of fuzzy set theory [13], aiming to provide a qualitative framework for handling uncertain information. Building upon Zadeh's work, Dubois and Prade formulated possibilistic logic [1] to integrate possibility theory into formal logic systems.

Possibilistic logic has been further developed for diverse applications such as default reasoning, belief revision, or decision-making under uncertainty with generalized approaches; see [6]. The goal of this paper is to introduce basic possibilistic logic with examples and visualizations. The focus is on providing examples with additional explanations and graphical representations. In this paper, we will focus solely on basic possibilistic logic. This work is structured as follows. In Section 2, basic possibilistic logic is explained, including its preliminaries for understanding the syntax and semantics of the logic. Before the core part of this paper, Section 3 compares possibilistic logic with classical logic, and Section 4 discusses its relationship with artificial intelligence. Section 5 illustrates the application of basic possibilistic logic through an example and graphical explanations. Finally, Section 6 summarizes and discusses the results of the previous sections.

2 Fundamentals of Basic Possibilistic Logic

The following introduction to basic possibilistic logic is inspired by the presentation of Dubois and Prade [2]. This section provides a brief overview of basic possibilistic logic. For more details, we refer to the works [2] and [1].

In this section, we first introduce the main concepts of possibility theory, which are necessary for understanding possibilistic logic. Then, we present basic possibilistic logic, covering its syntax, axioms, inference rules, and semantic aspects.

2.1 Possibility Theory

Possibility theory is explained in detail in [5] and [14]. It relies on the concept of possibility distributions.

Definition 1 *A possibility distribution is a function π that maps elements from a set of states U to a totally ordered scale $[0, 1]$, see [2]:*

$$\pi : U \rightarrow [0, 1].$$

The value $\pi(u)$ indicates the degree of possibility assigned to the state u . This function represents the agent's knowledge and helps differentiate between states that are more or less plausible. A state u is considered impossible if $\pi(u) = 0$, whereas it is fully possible if $\pi(u) = 1$.

Example 1 *Consider a weather forecasting scenario where an agent assesses the possibility of different weather conditions for tomorrow. Let $U = \{\text{sunny}, \text{cloudy}, \text{rainy}, \text{stormy}\}$ be the set of possible states. The agent assigns a possibility distribution π as follows:*

$$\pi(\text{sunny}) = 1, \quad \pi(\text{cloudy}) = 0.7, \quad \pi(\text{rainy}) = 0.3, \quad \pi(\text{stormy}) = 0.$$

The agent is completely certain that a sunny day is fully possible ($\pi(\text{sunny}) = 1$), considers cloudy weather strong possible ($\pi(\text{cloudy}) = 0.7$), finds rain unlikely ($\pi(\text{rainy}) = 0.3$), and states a storm impossible ($\pi(\text{stormy}) = 0$).

The basic measures of possibility theory model the possibility and necessity of events (see [2], [10]). If the occurrence of an event A is of interest, possibility theory provides two measures derived from the possibility distribution of A .

Definition 2 *The possibility of an event A is defined as:*

$$\Pi(A) = \sup_{u \in A} \pi(u).$$

The possibility measure represents the maximum possibility assigned to any state in A . It evaluates the extent to which A is consistent with π . Possibility measures satisfy the characteristic maxitivity property:

$$\Pi(A \cup B) = \max(\Pi(A), \Pi(B)).$$

Definition 3 *The necessity of an event A is defined as:*

$$N(A) = \inf_{u \notin A} (1 - \pi(u)).$$

The necessity measure represents the minimum of the inverted possibility values of all states not in A . It evaluates the extent to which A is certainly implied by π . Necessity measures satisfy the characteristic minimization property:

$$N(A \cap B) = \min(N(A), N(B)).$$

This property expresses that for $A \cap B$ to be certain, both A and B must be certain individually. The duality between these two measures is given by:

$$N(A) = 1 - \Pi(A^c),$$

where A^c is the complement of A .

Example 2 *Consider a possibility distribution π over a set of possible weather conditions: $U = \{\text{sunny}, \text{cloudy}, \text{rainy}\}$ with the values $\pi(\text{sunny}) = 1$, $\pi(\text{cloudy}) = 0.7$, and $\pi(\text{rainy}) = 0.3$.*

The possibility of the event "not rainy" $A = \{\text{sunny}, \text{cloudy}\}$ is given by:

$$\Pi(A) = \sup_{u \in A} \pi(u) = \max(\pi(\text{sunny}), \pi(\text{cloudy})) = \max(1, 0.7) = 1.$$

The necessity of "not rainy" is computed as:

$$N(A) = \inf_{u \notin A} (1 - \pi(u)) = 1 - \pi(\text{rainy}) = 1 - 0.3 = 0.7.$$

This means that the event "not rainy" is fully possible but not completely necessary.

2.2 Formal Foundations of Basic Possibilistic Logic

This section lays out the foundations of basic possibilistic logic. First, the syntax is introduced. Second, the semantic aspects of the logic are derived. Third, the inference rules are explained. Finally, the axioms of this logic are covered. Further references for this section are [2], [9].

Definition 4 *The syntax of a basic possibilistic logic formula is:*

$$(a, \alpha)$$

where a is a classical logic formula (e.g., $p \rightarrow q$) and α is the certainty level (the minimum necessity degree of a). The classical logic formula can be from propositional logic or first-order logic (see [2]).

The semantics of a possibilistic logic formula are based on the semantics of the corresponding classical logic formula. The pair (a, α) is interpreted as $N(a) > \alpha$, meaning that the formula a is at least certain to degree α . The higher the value of α , the stronger the belief in a .

Example 3 $(b \rightarrow f, 0.9)$ means that "birds fly" holds with certainty 0.9.

The following inference rules of basic possibilistic logic are necessary for the application of the logic in this paper. The *certainty weakening rule* states that if a formula holds with a certainty level α , it also holds with any lower certainty level β because we are being less demanding (see Equation 1).

$$\text{If } \beta \leq \alpha, \quad (a, \alpha) \vdash (a, \beta) \quad (1)$$

The *modus ponens rule* states that if $a \rightarrow b$ holds with certainty level α , and a also holds with certainty level α , then b must hold with certainty level α (see Equation 2).

$$(\neg a \vee b, \alpha), (a, \alpha) \vdash (b, \alpha), \quad \forall \alpha \in (0, 1] \quad (2)$$

In classical logic, the *resolution rule* allows us to combine two clauses that share a literal, eliminating that literal and inferring a new clause. In possibilistic logic, the resolution rule preserves the certainty level α since the inferred formula $b \vee c$ cannot be more certain than the least certain premise (see Equation 3).

$$(\neg a \vee b, \alpha), \quad (a \vee c, \alpha) \vdash (b \vee c, \alpha) \quad (3)$$

The *weakest link resolution rule* extends the resolution rule by considering different certainty levels α and β . It reflects the idea that the certainty of the conclusion cannot exceed the weakest certainty of the premises (see Equation 4).

$$(\neg a \vee b, \alpha), \quad (a \vee c, \beta) \vdash (b \vee c, \min(\alpha, \beta)) \quad (4)$$

For inference with basic possibilistic logic, a possibilistic logic base is used.

Definition 5 A *possibilistic knowledge base* is a collection of possibilistic formulas:

$$\Gamma = \{(a_i, \alpha_i), i = 1, \dots, m\} \quad (5)$$

Example 4 In a medical diagnosis system symptoms are associated with diseases with varying degrees of certainty. The possibilistic knowledge base is given by:

$$\Gamma = \{(Fever \Rightarrow Flu, 0.9), (Cough \Rightarrow Cold, 0.7), (Headache \Rightarrow Migraine, 0.8)\}.$$

This means that having a fever strongly suggests flu with a certainty level of 0.9, a cough suggests a cold with 0.7, and a headache leads to a migraine with 0.8.

The α -cut of a knowledge base contains only those formulas with a certainty level of at least α . It has a nested structure where Γ_α is a subset of Γ_β if $\alpha \geq \beta$.

Definition 6 The α -cut of a knowledge base is defined as:

$$\Gamma_\alpha = \{(a_i, \alpha_i) \in \Gamma \mid \alpha_i \geq \alpha\} \quad (6)$$

Example 5 Continuing with Example 4, the α -cut of the knowledge base for $\alpha = 0.8$ is:

$$\Gamma_\alpha = \{(Fever \Rightarrow Flu, 0.9), (Headache \Rightarrow Migraine, 0.8)\}.$$

A possibilistic inference (a, α) depends only on formulas with certainty levels $\geq \alpha$, as shown in Equation 7:

$$\Gamma \vdash (a, \alpha) \iff \Gamma_\alpha \vdash (a, \alpha) \quad (7)$$

An important property of knowledge bases is their inconsistency level.

Definition 7 The inconsistency level of a knowledge base Γ is defined as:

$$inc-l(\Gamma) = \max\{\alpha \mid \Gamma \vdash (\perp, \alpha)\} \quad (8)$$

where $\perp$ represents an inconsistent state in logic.

The inconsistency level represents the highest certainty level at which the knowledge base becomes inconsistent. Any formula with a certainty level higher than the inconsistency level remains free from inconsistency.

Example 6 In the above example, there are no conflicting statements (e.g., one rule stating “Fever implies Flu” and another stating “Fever does not imply Flu”). The inconsistency level $inc-l(\Gamma)$ is 0, indicating that the knowledge base is fully consistent at all certainty levels.

3 Differences from Classical Logic

Possibilistic logic extends classical logic by incorporating a certainty degree into classical logic formulas, see [2]. In classical logic, a statement is either true or false, corresponding to a binary value system. In contrast, in possibilistic logic, the certainty degree of a possibilistic formula is a non-binary value [1].

In classical logic, the presence of a contradiction makes the entire knowledge base invalid, allowing any formula to be derived. In difference possibilistic logic can handle inconsistency. The inconsistency degree of a knowledge base reflects the extent to which it is contradictory [2]. This feature makes possibilistic logic robust for reasoning in incomplete and inconsistent environments.

4 Relation to Artificial Intelligence

Possibilistic logic, with its strength in modeling certainty and handling inconsistency, has various application fields in artificial intelligence problems [2], [4]. The following application examples are only a glimpse into the possible fields. In default reasoning, possibilistic logic handles incomplete information reasoning. Additionally, possibilistic logic can be applied in information fusion, where knowledge is combined from different sources. The logic can also assist with decisions under uncertainty. One application is its use in medical diagnosis problems [7]. In [3], [11], connections between reasoning with possibilistic logic and machine learning are explored.

5 Applying Basic Possibilistic Logic

In the following, we will go through an example demonstrating the application of basic possibilistic logic. One of the classical examples in non-monotonic reasoning is the penguin problem. Basic possibilistic logic provides a structured way to handle exceptions to general rules, such as "birds fly," by associating rules with certainty levels and defining a preference ordering over possible worlds. Each possible world represents a combination of the language interpretation of the underlying rules, see [2]. This example can be found in a more concise form in [2], [12].

5.0.1 Default Rules

The knowledge base of this example consists of three default rules about birds and penguins. A default rule expresses general knowledge and holds in most cases, but it may have exceptions:

$$\begin{aligned} d_1 : b \rightarrow f & \quad (\text{"Birds fly"}), \\ d_2 : p \rightarrow \neg f & \quad (\text{"Penguins do not fly"}), \\ d_3 : p \rightarrow b & \quad (\text{"Penguins are birds"}). \end{aligned}$$

In the following, the goal is to assign certainty levels to these rules to obtain a possibilistic knowledge base. After the creation of the possibilistic knowledge base we are able to use it for reasoning. To construct this knowledge base, the following steps are necessary. First, the constraints from the rules are derived. Second, the possible worlds are ranked based on these constraints. Finally, using the ranked worlds, the certainty levels of the rules are determined.

5.0.2 Constraints

From the default rules, a set of comparative possibility constraints can be constructed. Comparative possibility constraints are derived from the idea that a rule $a \rightarrow b$ implies that situations where both a and b hold should be considered more possible than those where a holds but b does not, see [2]:

$$\begin{aligned} C_1 : b \wedge f & >_{\Pi} b \wedge \neg f & (\text{birds fly}), \\ C_2 : p \wedge \neg f & >_{\Pi} p \wedge f & (\text{penguins do not fly}), \\ C_3 : p \wedge b & >_{\Pi} p \wedge \neg b & (\text{penguins are birds}). \end{aligned}$$

For example, C_1 states that the worlds where birds fly are more possible than worlds where birds do not fly. These constraints establish an ordering over possible worlds, which will be examined next.

5.0.3 Possible Worlds

The propositional language for this example involves three propositions: b (bird), p (penguin), and f (flies). The set of possible worlds, Ω , consists of all possible combinations

of (b, f, p) represented by ω_i :

$$\Omega = \{\omega_0, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7\}.$$

Based on the constraints, these worlds are ranked to reflect which interpretations are more plausible.

In Table 1, each row represents a world with a unique combination of the three predi-

World	b (Bird)	f (Flies)	p (Penguin)	$d_1 : b \rightarrow f$	$d_2 : p \rightarrow \neg f$	$d_3 : p \rightarrow b$
ω_0	0	0	0			
ω_1	0	0	1		satisfied	violated
ω_2	0	1	0			
ω_3	0	1	1		violated	violated
ω_4	1	0	0	violated		
ω_5	1	0	1	violated	satisfied	satisfied
ω_6	1	1	0	satisfied		
ω_7	1	1	1	satisfied	violated	satisfied

Table 1: Truth values of bird (b), flies (f), and penguin (p) for each world ω_i , and whether each world satisfies or violates the default rules.

cates, along with an indication of whether the default rules are satisfied or violated. Notably, no world satisfies all rules, highlighting the core reasoning challenge. The worlds ω_0 and ω_2 neither satisfy nor violate any rule. The first default rule, $d_1 : b \rightarrow f$ ("Birds fly"), increases the plausibility of worlds where birds fly ($b \wedge f$) compared to those where birds do not fly ($b \wedge \neg f$). This rule influences the ranking between worlds ω_6, ω_7 (where the bird flies) and ω_4, ω_5 (where the bird does not fly). The second default rule, $d_2 : p \rightarrow \neg f$ ("Penguins do not fly"), prioritizes worlds where penguins do not fly ($p \wedge \neg f$) over those where penguins fly ($p \wedge f$). This affects the relative plausibility of ω_5, ω_1 (where the penguin does not fly) compared to ω_7, ω_3 (where the penguin flies). The third default rule, $d_3 : p \rightarrow b$ ("Penguins are birds"), ranks the worlds ω_5, ω_7 higher than ω_1, ω_3 . By incorporating these rules, we derive the following constraints on possibility degrees:

$$\begin{aligned} C_1 : \max(\pi(\omega_6), \pi(\omega_7)) &> \max(\pi(\omega_4), \pi(\omega_5)), \\ C_2 : \max(\pi(\omega_5), \pi(\omega_1)) &> \max(\pi(\omega_3), \pi(\omega_7)), \\ C_3 : \max(\pi(\omega_5), \pi(\omega_7)) &> \max(\pi(\omega_1), \pi(\omega_3)). \end{aligned}$$

These constraints guide the ranking of possible worlds according to their plausibility in the given possibilistic logic framework.

5.0.4 Ranking of the Possible Worlds

By combining the constraints, we can determine an ordering of the worlds where no constraint is violated. The graph in Figure 1 illustrates the possibility relations between the worlds.

For example, constraint C_1 states that either ω_6 (blue arrows) or ω_7 (red arrows) must be more possible than ω_4 and ω_5 . The correct ordering is achieved if the worlds can be arranged into levels while ensuring that the arrows only point downward from a higher possibility level to a lower one. In this example, the ordering is color-coded into groups: blue (C_1), orange (C_2), and yellow (C_3). By following the constraints, we determine the

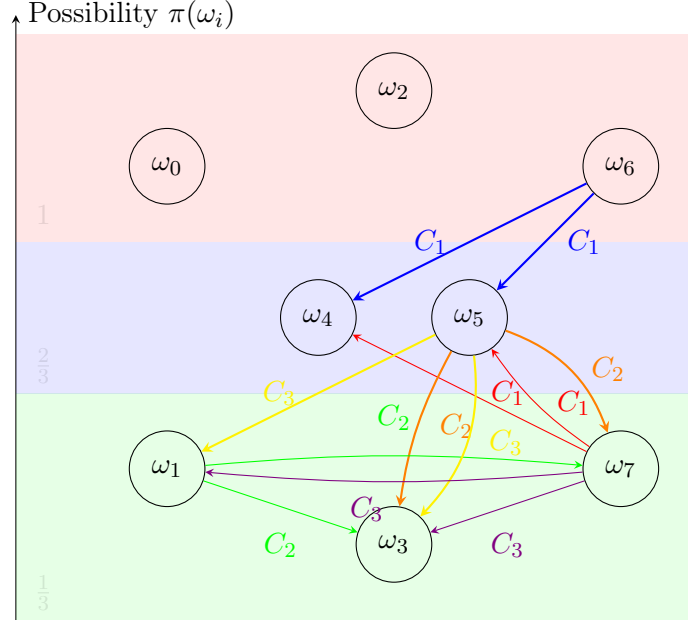


Figure 1: Possibility of the worlds ω_i and possibility relations between the worlds, represented by the colored arrows.

ranking of the worlds as follows:

$$\begin{aligned}\pi(\omega_0) &= \pi(\omega_2) = \pi(\omega_6) = 1, \\ \pi(\omega_4) &= \pi(\omega_5) = \frac{2}{3}, \\ \pi(\omega_1) &= \pi(\omega_3) = \pi(\omega_7) = \frac{1}{3}.\end{aligned}$$

In this example, we use numerical values to represent the different possibility levels of the worlds. The worlds ω_0 and ω_2 hold the highest possibility level since no constraints downgrade them.

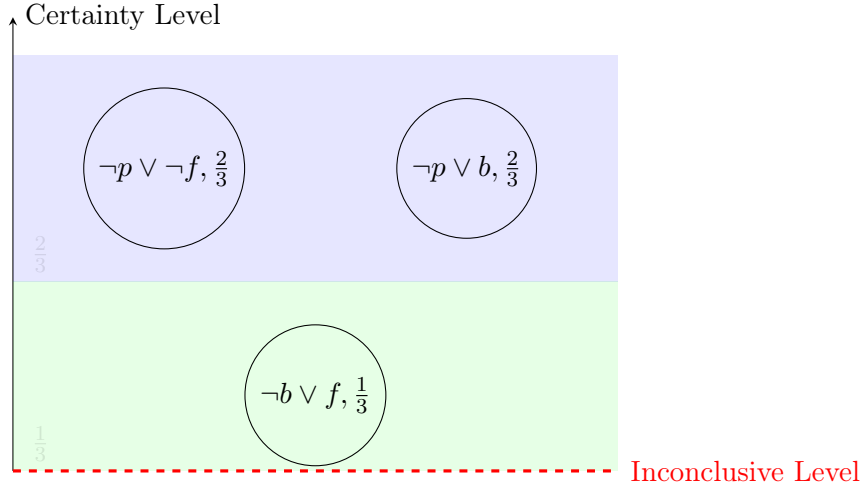


Figure 2: Possibilistic Knowledge Base of the Penguin Problem with the elements arranged over the certainty

5.0.5 Creating Knowledge Base

We calculate the necessity degrees from the possibility of the worlds as follows with the duality relationship of possibility and necessity:

$$\begin{aligned}
 N(\neg p \vee \neg f) &= \min\{1 - \pi(\omega) \mid \omega \models p \wedge f\} \\
 &= \min(1 - \pi(\omega_3), 1 - \pi(\omega_7)) = \frac{2}{3} \\
 N(\neg b \vee f) &= \min\{1 - \pi(\omega) \mid \omega \models b \wedge \neg f\} \\
 &= \min(1 - \pi(\omega_4), 1 - \pi(\omega_5)) = \frac{1}{3} \\
 N(\neg p \vee b) &= \min\{1 - \pi(\omega) \mid \omega \models p \wedge \neg b\} \\
 &= \min(1 - \pi(\omega_1), 1 - \pi(\omega_3)) = \frac{2}{3}
 \end{aligned}$$

Thus, we obtain the possibilistic knowledge base:

$$K = \{(\neg p \vee \neg f, \frac{2}{3}), (\neg p \vee b, \frac{2}{3}), (\neg b \vee f, \frac{1}{3})\}.$$

The possibilistic knowledge base is visualized in Figure 2.

5.0.6 Reasoning with Uncertain Information

In this section, we determine whether Tweety flies by introducing relevant information about itself. First, we introduce that Tweety is a bird:

$$F = \{(b, 1)\} \quad (\text{Tweety is a bird with certainty } 1).$$

Using the rule $(\neg b \vee f, 1/3)$ from the knowledge base K , we apply the weakest link resolution rule (Equation 4):

$$(\neg b \vee f, 1/3) \quad \text{and} \quad (b, 1) \vdash (f, \min(1, 1/3)) = (f, 1/3).$$

Thus, we conclude that Tweety flies with a certainty level of $\frac{1}{3}$. The result is illustrated graphically in Figure 3. The violet arrow symbolizes the inference using the weakest link resolution rule.

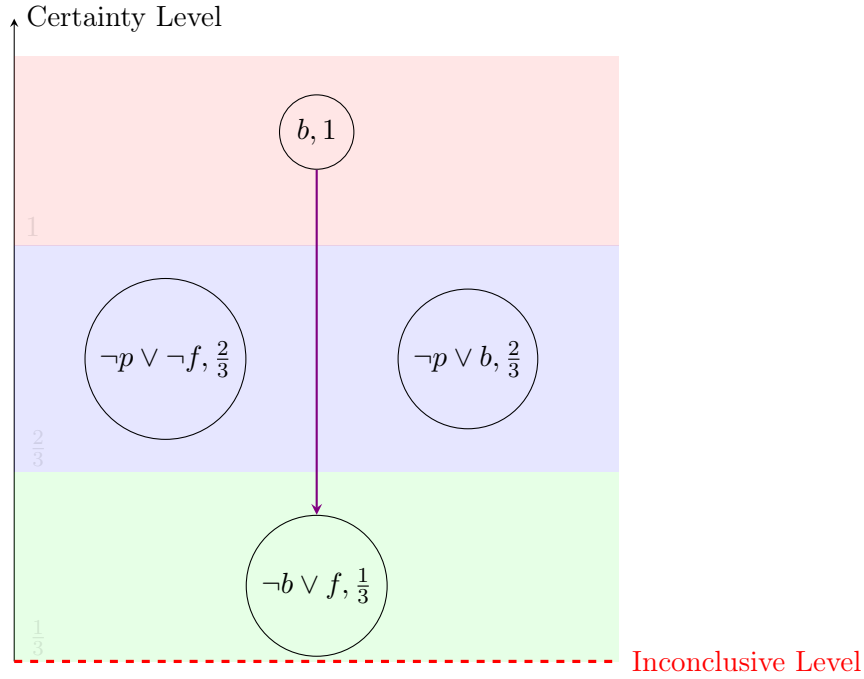


Figure 3: Possibilistic Knowledge Base of the Penguin Problem with first information about Tweety

If we introduce new information stating that Tweety is also a penguin:

$$F = \{(b, 1), (p, 1)\} \quad (\text{Tweety is a penguin with certainty } 1),$$

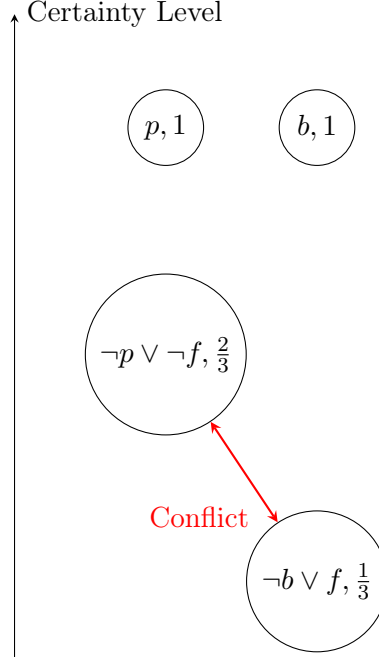


Figure 4: Possibilistic Knowledge Base of the Penguin Problem with new information about Tweety

then, using $(\neg p \vee \neg f, 2/3)$, we derive:

$$(\neg p \vee \neg f, 2/3) \quad \text{and} \quad (p, 1) \vdash (\neg f, \min(1, 2/3)) = (\neg f, 2/3).$$

Since this results in conflicting conclusions:

$$(\neg f, 2/3) \quad \text{and} \quad (f, 1/3) \vdash (\perp, \min(2/3, 1/3)) = (\perp, 1/3),$$

the system detects an inconsistency of level $\frac{1}{3}$. Given the principle that higher certainty information takes precedence, the inference that Tweety flies is overridden, leading to the final conclusion:

$$K \cup F \vdash (\neg f, 2/3),$$

which means "Tweety does not fly" with a certainty of $\frac{2}{3}$. The results are illustrated in Figure 4. It can be seen that the inconclusive level is raised to a level where no conflict exists between the information.

This example shows how possibilistic logic can handle inconsistency when new information is added. The calculation of the necessities of the rules out of the possible worlds is one key aspect in this example.

6 Discussion and Conclusion

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